Mud card

- **going through the timeline of decision tree algos was really helpful! was wondering whether in x gradient boost we throw away the first few trees because the later trees would've (presumably) addressed more errors in previous trees.**
  - No, all trees are kept in XGBoost and in Gradient Boosting as well
  - Trees are not thrown out because every tree is part of the model and essential to the final prediction
  - Consider that the trees are not independent.
  - The next tree tries to improve the previous tree by predicting its residuals
  - If you throw away the early trees, the residuals completely change
- **Since XGBoost handles missing values, do we still need to impute NaNs for categorical and ordinal features and change NaNs to a string value for it to then be treated as another category by OneHotEncoder/OrdinalEncoder?**
  - Yep, that's still recommended.
- **What is the imputation method that mentioned in the last part? Where can we find more info about it?**
  - That was the iterative imputer or multivariate imputation we covered in lecture 18
- **Will the exam potentially have combinatorics questions like the handful we saw in class?**
  - Yep
  - Generally speaking, the goal of the exam is not to check how well you can memorize stuff
  - The goal is to check whether you understand the concepts and can apply the methods you learnt

The supervised ML pipeline

0. Data collection/manipulation: you might have multiple data sources and/or you might have more data than you need

- you need to be able to read in datasets from various sources (like csv, excel, SQL, parquet, etc)
- you need to be able to filter the columns/rows you need for your ML model
- you need to be able to combine the datasets into one dataframe

1. Exploratory Data Analysis (EDA): you need to understand your data and verify that it doesn't contain errors

- do as much EDA as you can!

2. Split the data into different sets: most often the sets are train, validation, and test (or holdout)

- practitioners often make errors in this step!
- you can split the data randomly, based on groups, based on time, or any other non-standard way if necessary to answer your ML question

3. Preprocess the data: ML models only work if X and Y are numbers! Some ML models additionally require each feature to have 0 mean and 1 standard deviation (standardized features)

- often the original features you get contain strings (for example a gender feature would contain 'male', 'female', 'non-binary', 'unknown') which needs to be transformed into numbers
- often the features are not standardized (e.g., age is between 0 and 100) but it needs to be standardized

4. Choose an evaluation metric: depends on the priorities of the stakeholders

- often requires quite a bit of thinking and ethical considerations

5. Choose one or more ML techniques: it is highly recommended that you try multiple models

- start with simple models like linear or logistic regression
- try also more complex models like nearest neighbors, support vector machines, random forest, etc.

6. Tune the hyperparameters of your ML models (aka cross-validation or hyperparameter tuning)

- ML techniques have hyperparameters that you need to optimize to achieve best performance
- for each ML model, decide which parameters to tune and what values to try
- loop through each parameter combination
  - train one model for each parameter combination
  - evaluate how well the model performs on the validation set
- take the parameter combo that gives the best validation score
- evaluate that model on the test set to report how well the model is expected to perform on previously unseen data

7. Interpret your model: black boxes are often not useful

- check if your model uses features that make sense (excellent tool for debugging)

- often model predictions are not enough, you need to be able to explain how the model arrived to a particular prediction (e.g., in health care)

Global feature importance metrics

By the end of this module, you will be able to

- perform permutation feature importance calculation
- study the coefficients of linear models
- outlook to other metrics

Motivation

- debugging ML models is tough
  - a model that runs without errors/warning is not necessarily correct
- how do you know that your model is correct?
  - check test set predictions
    - in regression: check points with a large difference between true and predicted values
    - in classification: confusion matrix, check out FPs and FNs
  - inspect your model
    - especially useful for non-linear models
    - metrics to measure how much a model depends on a feature is one way to inspect your model

Global feature importance metrics

By the end of this module, you will be able to

- **perform permutation feature importance calculation**
- study coefficients of linear models
- outlook to other metrics

Permutation feature importance

- model agnostic, you can use it with any supervised ML model
- steps:
  - train a model and calculate a test score :)
  - randomly shuffle a single feature in the test set
  - recalculate the test score with the shuffled data

  - model score worsens because the shuffling breaks the relationship between feature and target
  - the larger the difference, the more important the feature is

In [1]:
```
import numpy as np
import pandas as pd
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.pipeline import make_pipeline
from sklearn.model_selection import GridSearchCV
from sklearn.model_selection import train_test_split
from sklearn.model_selection import StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder
import matplotlib.pylab as plt

df = pd.read_csv('../data/adult_data.csv')
label = 'gross-income'
y = df[label]
df.drop(columns=[label],inplace=True)
X = df
ftr_names = X.columns
print(X.head())
print(y)
```

```
   age         workclass  fnlwgt  education  education-num  \
0   39         State-gov   77516  Bachelors             13
1   50  Self-emp-not-inc   83311  Bachelors             13
2   38           Private  215646    HS-grad              9
3   53           Private  234721       11th              7
4   28           Private  338409  Bachelors             13

       marital-status         occupation   relationship   race     sex
\
0       Never-married       Adm-clerical  Not-in-family  White    Male
1  Married-civ-spouse    Exec-managerial        Husband  White    Male
2            Divorced  Handlers-cleaners  Not-in-family  White    Male
3  Married-civ-spouse  Handlers-cleaners        Husband  Black    Male
4  Married-civ-spouse     Prof-specialty           Wife  Black  Female

   capital-gain  capital-loss  hours-per-week  native-country
0          2174             0              40   United-States
1             0             0              13   United-States
2             0             0              40   United-States
3             0             0              40   United-States
4             0             0              40            Cuba
0        <=50K
1        <=50K
2        <=50K
3        <=50K
4        <=50K
         ...
32556    <=50K
32557     >50K
32558    <=50K
32559    <=50K
32560     >50K
Name: gross-income, Length: 32561, dtype: object
```

In [2]:

```python
def ML_pipeline_kfold(X,y,random_state,n_folds):
    # create a test set
    X_other, X_test, y_other, y_test = train_test_split(X, y, test_size=0.2,
    # splitter for _other
    kf = StratifiedKFold(n_splits=n_folds,shuffle=True,random_state=random_s
    # create the pipeline: preprocessor + supervised ML method
    cat_ftrs = ['workclass','education','marital-status','occupation','relat
    cont_ftrs = ['age','fnlwgt','education-num','capital-gain','capital-loss
    # one-hot encoder
    categorical_transformer = Pipeline(steps=[
        ('onehot', OneHotEncoder(sparse_output=False,handle_unknown='ignore'
    # standard scaler
    numeric_transformer = Pipeline(steps=[
        ('scaler', StandardScaler())])
    preprocessor = ColumnTransformer(
        transformers=[
            ('num', numeric_transformer, cont_ftrs),
            ('cat', categorical_transformer, cat_ftrs)])
    pipe = make_pipeline(preprocessor,SVC())
    #pipe = make_pipeline(preprocessor,RandomForestClassifier())
    # the parameter(s) we want to tune
    param_grid = {'svc__C': [0.01, 0.1, 1, 10, 100],
```

```
                        'svc__gamma': [0.01, 0.1, 1, 10, 100]}

    # prepare gridsearch
    grid = GridSearchCV(pipe, param_grid=param_grid,cv=kf, return_train_scor
    # do kfold CV on _other
    grid.fit(X_other, y_other)
    return grid, X_test, y_test
```

Be careful, SVM is used on a relatively large dataset

In [3]:
```
model, X_test, y_test = ML_pipeline_kfold(X,y,42,4)
print(model.best_score_)
print(model.score(X_test,y_test))
print(model.best_params_)

# save the output so I can use it later
import pickle
file = open('../results/grid.save', 'wb')
pickle.dump((model,X_test,y_test),file)
file.close()
```

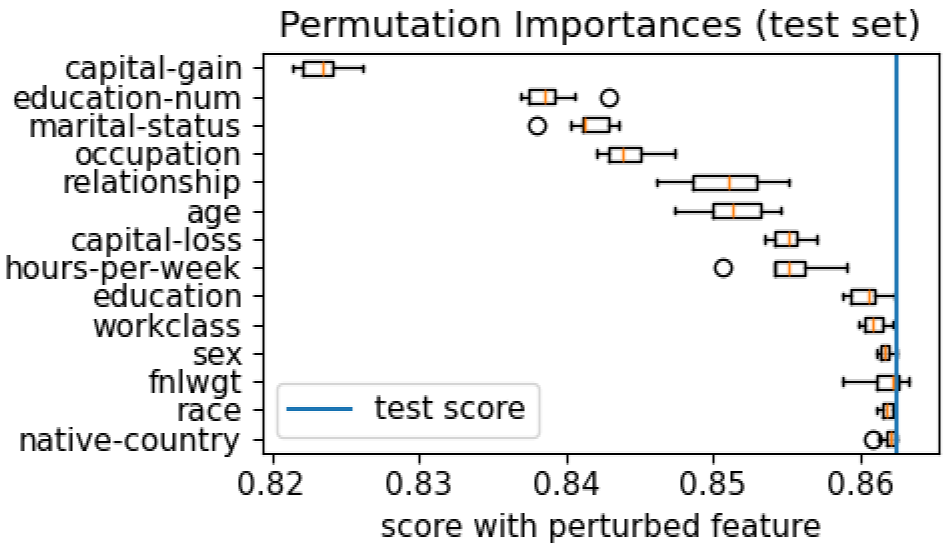
```
Fitting 4 folds for each of 25 candidates, totalling 100 fits
0.8545377764127764
0.8624289881774911
{'svc__C': 1, 'svc__gamma': 0.1}
```

In [3]:
```
import pickle
file = open('../results/grid.save', 'rb')
model, X_test, y_test = pickle.load(file)
file.close()

np.random.seed(42)

nr_runs = 10
scores = np.zeros([len(ftr_names),nr_runs])

test_score = model.score(X_test,y_test)
print('test score = ',np.around(test_score,3))
print('test baseline = ',np.around(np.sum(y_test == ' <=50K')/len(y_test),3)
# loop through the features
for i in range(len(ftr_names)):
    print('shuffling '+str(ftr_names[i]))
    acc_scores = []
    for j in range(nr_runs):
        X_test_shuffled = X_test.copy()
        X_test_shuffled[ftr_names[i]] = np.random.permutation(X_test[ftr_nam
        acc_scores.append(model.score(X_test_shuffled,y_test))
    print('   shuffled test score:',np.around(np.mean(acc_scores),3),'+/-',r
    scores[i] = acc_scores
```

```
test score =  0.862
test baseline =  0.759
shuffling age
   shuffled test score: 0.851 +/- 0.002
shuffling workclass
   shuffled test score: 0.861 +/- 0.001
shuffling fnlwgt
   shuffled test score: 0.862 +/- 0.001
shuffling education
   shuffled test score: 0.86 +/- 0.001
shuffling education-num
   shuffled test score: 0.839 +/- 0.002
shuffling marital-status
   shuffled test score: 0.842 +/- 0.002
shuffling occupation
   shuffled test score: 0.844 +/- 0.002
shuffling relationship
   shuffled test score: 0.851 +/- 0.003
shuffling race
   shuffled test score: 0.862 +/- 0.0
shuffling sex
   shuffled test score: 0.862 +/- 0.0
shuffling capital-gain
   shuffled test score: 0.823 +/- 0.001
shuffling capital-loss
   shuffled test score: 0.855 +/- 0.001
shuffling hours-per-week
   shuffled test score: 0.855 +/- 0.002
shuffling native-country
   shuffled test score: 0.862 +/- 0.001
```

In [4]:

```
sorted_indcs = np.argsort(np.mean(scores,axis=1))[::-1]
plt.rcParams.update({'font.size': 11})
plt.figure(figsize=(5,3))
plt.boxplot(scores[sorted_indcs].T,tick_labels=ftr_names[sorted_indcs],vert=
plt.axvline(test_score,label='test score')
plt.title("Permutation Importances (test set)")
plt.xlabel('score with perturbed feature')
plt.legend()
plt.tight_layout()
plt.show()
```



Check out sklearn's permutation importance!

https://scikit-learn.org/stable/modules/generated/sklearn.inspection.permutation_importance.html

https://scikit-learn.org/stable/modules/permutation_importance.html#permutation-importance

Cons of permutation feature importance

- strongly correlated features
  - if one of the features is shuffled, the model can still use the other correlated feature
  - both features appear to be less important than they actually are
  - solution:
    - check the correlation matrix plot
    - remove all but one of the strongly correlated features
- no feature interactions
  - one feature might appear unimportant but combined with another feature could be important
  - solution:
    - permute two features to measure how important feature pairs are
    - this can be computationally expensive

Quiz

Global feature importance metrics

By the end of this module, you will be able to

- perform permutation feature importance calculation
- **study the coefficients of linear models**
- outlook to other metrics

Coefficients of linear models

- the coefficients of linear and logistic regression can be used as a measure of feature importance **ONLY IF** all features have the same mean (usually 0) and the same standard deviation (usually 1)
  - all features meaning that the one-hot encoded and ordinal features as well!
- then the absolute value of the coefficients can be used to rank them

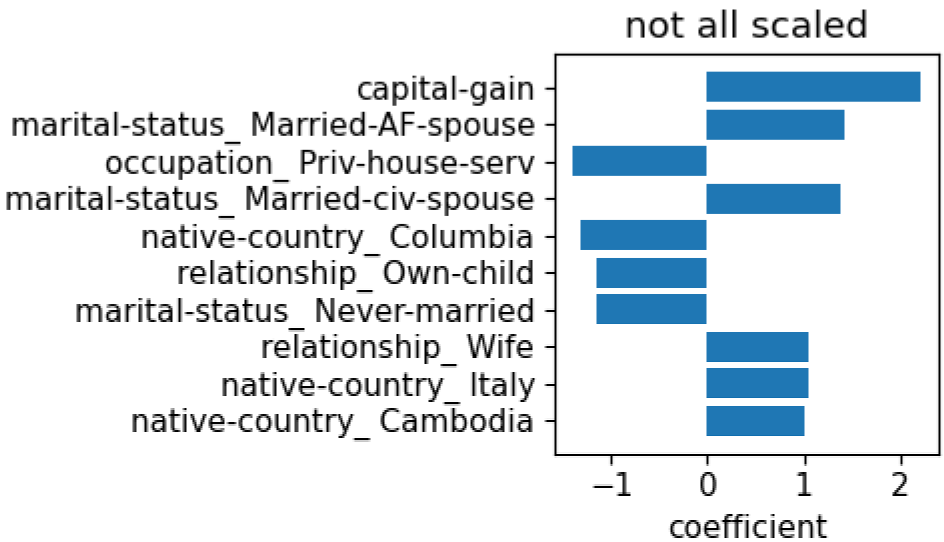
Let's rewrite the kfold CV function a bit

In [5]:
```
from sklearn.linear_model import LogisticRegression
def ML_pipeline_kfold_LR1(X,y,random_state,n_folds):
    # create a test set
    X_other, X_test, y_other, y_test = train_test_split(X, y, test_size=0.2,
    # splitter for _other
    kf = StratifiedKFold(n_splits=n_folds,shuffle=True,random_state=random_s
    # create the pipeline: preprocessor + supervised ML method
    cat_ftrs = ['workclass','education','marital-status','occupation','relat
    cont_ftrs = ['age','fnlwgt','education-num','capital-gain','capital-loss
    # one-hot encoder
    categorical_transformer = Pipeline(steps=[
        ('onehot', OneHotEncoder(sparse_output=False,handle_unknown='ignore'
    # standard scaler
    numeric_transformer = Pipeline(steps=[
        ('scaler', StandardScaler())])
    preprocessor = ColumnTransformer(
        transformers=[
            ('num', numeric_transformer, cont_ftrs),
            ('cat', categorical_transformer, cat_ftrs)])
    pipe = make_pipeline(preprocessor,LogisticRegression(penalty='l2',solver
    # the parameter(s) we want to tune
    param_grid = {'logisticregression__C': [0.01, 0.1, 1, 10,100]}
    # prepare gridsearch
    grid = GridSearchCV(pipe, param_grid=param_grid,cv=kf, return_train_scor
    # do kfold CV on _other
    grid.fit(X_other, y_other)
    feature_names = cont_ftrs + \
                list(grid.best_estimator_[0].named_transformers_['cat'][0].g
    return grid, np.array(feature_names), X_test, y_test
```

In [6]:
```
grid, feature_names, X_test, y_test = ML_pipeline_kfold_LR1(X,y,42,4)
print('test score:',grid.score(X_test,y_test))
coefs = grid.best_estimator_[-1].coef_[0]
```

```
sorted_indcs = np.argsort(np.abs(coefs))

plt.figure(figsize=(5,3))
plt.rcParams.update({'font.size': 11})
plt.barh(np.arange(10),coefs[sorted_indcs[-10:]])
plt.yticks(np.arange(10),feature_names[sorted_indcs[-10:]])
plt.xlabel('coefficient')
plt.title('not all scaled')
plt.tight_layout()
plt.savefig('../figures/LR_coefs_notscaled.png',dpi=300)
plt.show()
```

```
test score: 0.8581298940580377
```

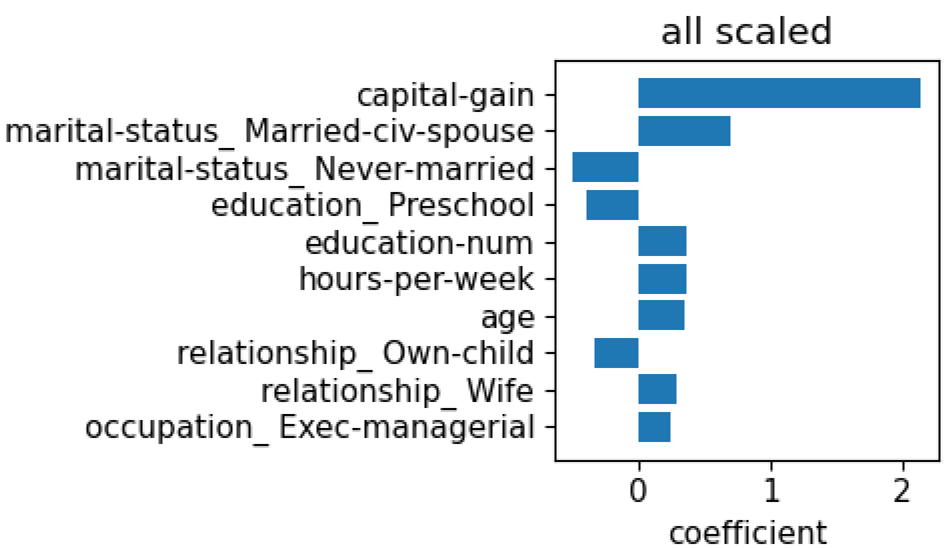


In [7]:
```
from sklearn.linear_model import LogisticRegression
def ML_pipeline_kfold_LR2(X,y,random_state,n_folds):
    # create a test set
    X_other, X_test, y_other, y_test = train_test_split(X, y, test_size=0.2,
    # splitter for _other
    kf = StratifiedKFold(n_splits=n_folds,shuffle=True,random_state=random_s
    # create the pipeline: preprocessor + supervised ML method
    cat_ftrs = ['workclass','education','marital-status','occupation','relat
    cont_ftrs = ['age','fnlwgt','education-num','capital-gain','capital-loss
    # one-hot encoder
    categorical_transformer = Pipeline(steps=[
        ('onehot', OneHotEncoder(sparse_output=False,handle_unknown='ignore'
    # standard scaler
    numeric_transformer = Pipeline(steps=[
        ('scaler', StandardScaler())])
    preprocessor = ColumnTransformer(
        transformers=[
            ('num', numeric_transformer, cont_ftrs),
            ('cat', categorical_transformer, cat_ftrs)])
    final_scaler = StandardScaler()
    pipe = make_pipeline(preprocessor,final_scaler,LogisticRegression(penalt
    # the parameter(s) we want to tune
    param_grid = {'logisticregression__C': [0.01, 0.1, 1, 10,100]}
    # prepare gridsearch
    grid = GridSearchCV(pipe, param_grid=param_grid,cv=kf, return_train_scor
```

```
    # do kfold CV on _other
    grid.fit(X_other, y_other)
    feature_names = cont_ftrs + \
                list(grid.best_estimator_[0].named_transformers_['cat'][0].g

    return grid, np.array(feature_names), X_test, y_test
```

In [8]:
```
grid, feature_names, X_test, y_test = ML_pipeline_kfold_LR2(X,y,42,4)
print('test score:',grid.score(X_test,y_test))
coefs = grid.best_estimator_[-1].coef_[0]
sorted_indcs = np.argsort(np.abs(coefs))

plt.figure(figsize=(5,3))
plt.rcParams.update({'font.size': 11})
plt.barh(np.arange(10),coefs[sorted_indcs[-10:]])
plt.yticks(np.arange(10),feature_names[sorted_indcs[-10:]])
plt.xlabel('coefficient')
plt.title('all scaled')
plt.tight_layout()
plt.savefig('../figures/LR_coefs_scaled.png',dpi=300)
plt.show()
```

```
test score: 0.857976354982343
```



Drawing Drawing

Global feature importance metrics

By the end of this module, you will be able to

- perform permutation feature importance calculation
- study the coefficients of linear models
- **outlook to other metrics**

- SVM:

    - SVC.coef_ and SVR.coef_ can be used as a metric of feature importance if **all** features are standardized
    - for linear SVMs only!
- random forest:
    - RandomForestRegressor.feature_importances_ and RandomForestClassification.feature_importances_
    - gini importance or mean decrease impurity, see here and here
- XGBoost:
    - five different metrics are implemented, see here and here

Quiz

Mudcard

```
In [ ]:
```